

Erdős Problem #653

EP-653

L1 - TRACTABLE

OPEN

Geometry

ORIGINAL PROBLEM STATEMENT

Let $x_1, \dots, x_n \in \mathbb{R}^2$ and let $R(x_i) = \#\{|x_j - x_i| : j \neq i\}$, where the points are ordered such that $R(x_1) \leq \dots \leq R(x_n)$. Let $g(n)$ be the maximum number of distinct values the $R(x_i)$ can take. Is it true that $g(n) \geq (1 - o(1))n$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>